Amazon S3

Simple Storage Service — Comprehensive Study Notes

AWS Foundational Storage April 4, 2026 Production-Grade Reference

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SECTION 01

Introduction

Amazon Simple Storage Service (S3) is the foundation of AWS storage, launched in 2006 as one of the first AWS services. It revolutionized storage by providing unlimited, durable, highly available object storage accessible via simple HTTP APIs. Today, S3 stores **trillions of objects** and regularly delivers millions of requests per second, serving as the backbone for countless applications, data lakes, backup systems, content distribution networks, and machine learning pipelines.

S3's simplicity masks its sophistication. Behind the straightforward "upload and retrieve" interface lies a powerful system with **11 nines of durability** (99.999999999%), multiple storage classes optimized for different access patterns, intelligent lifecycle management, versioning for data protection, encryption options, fine-grained access controls, event notifications, replication across regions, and analytics capabilities.

COST IMPLICATIONS

Choosing the wrong storage class can waste thousands of dollars monthly. Improper lifecycle policies can delete critical data or fail to reduce costs. Public bucket exposures have led to massive data breaches. Yet with proper configuration, S3 provides incredibly cost-effective storage — as low as

$0.99 PER TB/MONTH

for Glacier Deep Archive.

SECTION 02

S3 Fundamentals

Object Storage Model

S3 is *object storage*, not file or block storage. Each object consists of:

1. **Key** — Unique identifier (like filename with path)
2. **Value** — Object data (0 bytes to 5 TB)
3. **Metadata** — System and user-defined metadata
4. **Version ID** — Unique identifier if versioning enabled
5. **Access Control** — Permissions for the object

Bucket Structure

```
Bucket: my-bucket  (globally unique name)
├── Object: documents/report.pdf
│   ├── Value:      [binary data]
│   ├── Metadata:   content-type=application/pdf, custom-tag=finance
│   └── Version ID: abc123xyz
└── Object: images/logo.png
```

Key Characteristics

- **Unlimited Storage** — No capacity limits
- **Durability** — 99.999999999% (11 9s) — lose 1 object per 10 million over 10 million years
- **Availability** — 99.9% to 99.99% depending on storage class

- **Scalability** — Scales automatically, handles millions of requests per second
- **Global Namespace** — Bucket names must be globally unique
- **Regional Service** — Buckets created in specific region; data doesn't leave region unless configured

S3 vs. File Systems

Feature	S3 (Object Storage)	File System (Block/File)
Structure	Flat namespace, objects with keys	Hierarchical directories
Access	HTTP/HTTPS APIs	POSIX file operations
Modification	Entire object replaced	Partial file updates
Performance	Optimized for sequential access	Random I/O supported
Consistency	Strong consistency	Strong consistency
Use Case	Backups, media, data lakes	Databases, applications

SECTION 03

Storage Classes

S3 offers multiple storage classes optimized for different access patterns and costs. Choosing the right class is one of the most impactful cost decisions in AWS.

S3 Standard

Frequently accessed data

Availability: 99.99%

Retrieval fee: None

Min duration: None

$0.023/GB/mo

Intelligent-Tiering

Unknown/changing access patterns

Availability: 99.9%

Auto-moves between tiers

Monitoring: $0.0025/1k objects

$0.023/GB/mo*

Standard-IA

Infrequent but rapid access

Availability: 99.9%

Retrieval: $0.01/GB

Min duration: 30 days

$0.0125/GB/mo

One Zone-IA

Infrequent, recreatable data

Availability: 99.5%

Risk: Lost if AZ destroyed

Min duration: 30 days

$0.01/GB/mo

Glacier Instant

Archive with instant access (ms)

Availability: 99.9%

Retrieval: $0.03/GB

Min duration: 90 days

$0.004/GB/mo

Glacier Flexible

Archive — minutes to hours

Expedited: 1-5 min ($0.03/GB)

Standard: 3-5 hr ($0.01/GB)

Min duration: 90 days

$0.0036/GB/mo

Glacier Deep Archive

Lowest cost, long-term

Standard: 12 hr ($0.02/GB)

Bulk: 48 hr ($0.0025/GB)

Min duration: 180 days

$0.00099/GB/mo

Storage Class Comparison

Class	$/GB/Month	Retrieval Fee	Min Duration	Use Case
Standard	$0.023	**None**	None	Hot data
Intelligent-Tiering	$0.023*	**None***	None	Unknown patterns
Standard-IA	$0.0125	$0.01/GB	30 days	Cool data
One Zone-IA	$0.01	$0.01/GB	30 days	Recreatable cool
Glacier Instant	$0.004	$0.03/GB	90 days	Archive instant
Glacier Flexible	$0.0036	$0.01/GB	90 days	Archive hours
Glacier Deep Archive	**$0.00099**	$0.02/GB	180 days	Archive days

* Intelligent-Tiering charges vary by tier accessed. Prices shown for us-east-1.

SECTION 04

Versioning

Versioning preserves, retrieves, and restores every version of every object stored in your bucket.

States

- **Unversioned (default)** — Objects have null version ID
- **Versioning-enabled** — New versions created on update
- **Versioning-suspended** — No new versions, but existing preserved

How It Works

```
Upload report.pdf  → Version ID: v1  (latest)
Upload report.pdf  → Version ID: v2  (latest),  v1 (older)
Upload report.pdf  → Version ID: v3  (latest),  v2, v1 (older)
Delete report.pdf  → Delete marker   (latest),  v3, v2, v1 still exist
```

Benefits

- Protect against accidental deletion
- Recover from application failures
- Archive previous versions for auditing
- Meet compliance requirements

CONSIDERATIONS

Storage costs multiply since each version is stored separately. Use lifecycle policies to manage old versions. Versioning **CANNOT BE DISABLED ONCE ENABLED** — it can only be suspended.

SECTION 05

Lifecycle Policies

Automate transitioning objects between storage classes or deleting them to reduce cost.

Lifecycle Actions

1. **Transition Actions** — Move to cheaper storage class over time
2. **Expiration Actions** — Permanently delete objects

Transition Constraints

From Class	To Class	Min Days
Standard	Standard-IA	30
Standard	Intelligent-Tiering	0
Standard	Glacier Instant	0
Standard-IA	Glacier Flexible	30
Glacier Instant	Glacier Deep Archive	90

CANNOT TRANSITION

You cannot transition FROM any class *back* to Standard. Objects smaller than 128 KB cannot be transitioned to IA classes (except Intelligent-Tiering).

SECTION 06

Encryption

S3 provides multiple encryption options for data at rest and in transit.

Server-Side Encryption (SSE)

- **SSE-S3 (Default)** — S3-managed keys, AES-256, free, enabled by default on new buckets
- **SSE-KMS** — AWS KMS managed keys, CloudTrail audit trail, key rotation, cost: ~$0.03 per 10,000 API calls

- **SSE-C** — Customer-provided keys, key not stored by AWS, must provide key with each request

Client-Side Encryption

- Encrypt before uploading and decrypt after downloading
- You manage everything — full control over key material

Encryption in Transit

- HTTPS enforced via bucket policy
- TLS 1.2+ supported

Encryption Comparison

Method	Key Management	Audit	Cost	Use Case
SSE-S3	AWS	No	Free	Default, simple
SSE-KMS	AWS KMS	CloudTrail	API calls	Compliance, audit
SSE-C	Customer	No	Free	Full control
Client-Side	Customer	No	Free	Complete control

SECTION 07

S3 Access Control

Multiple layered mechanisms control access to S3 resources.

1. **IAM Policies** — Control what AWS users/roles can do
2. **Bucket Policies** — Control access to entire bucket or objects
3. **Access Control Lists (ACLs)** — Legacy mechanism (AWS recommends using policies instead)
4. **Block Public Access** — Account/bucket-level settings to prevent public exposure
5. **Access Points** — Simplify managing access for shared datasets

Access Evaluation Order

1. Check Block Public Access settings
2. Evaluate IAM policies (user/role)
3. Evaluate bucket policies
4. Evaluate ACLs
5. Evaluate Access Point policies

> **EVALUATION RULE**
>
> The result is the union of all ALLOW permissions. However, any explicit **DENY** wins and overrides any ALLOW.

SECTION 08

S3 Performance

Request Rate Performance

- Prefix-based partitioning: **3,500 PUT/COPY/POST/DELETE** and **5,500 GET/HEAD** requests per second per prefix
- Multiple prefixes provide linear scalability

```
my-bucket/prefix1/  →  5,500 GET/s
my-bucket/prefix2/  →  5,500 GET/s
my-bucket/prefix3/  →  5,500 GET/s
──────────────────────────────────
Total               → 16,500 GET/s
```

Multipart Upload

For objects > 100 MB, use multipart upload:

- Upload parts in parallel for improved throughput
- Quick recovery from network failures
- **Required** for objects > 5 GB

Transfer Acceleration

Enable faster uploads/downloads via CloudFront edge locations — typical speedup of **50–500%** depending on geographic location.

Performance Optimization Techniques

Technique	Use Case	Performance Gain
Multipart Upload	Objects > 100 MB	**10x+ throughput**
Transfer Acceleration	Global uploads	**50–500% faster**
CloudFront	Frequent reads	**<50ms globally**
S3 Select	Query CSV/JSON	**80% cost reduction**
Byte-Range Fetches	Large files	Parallel downloads

SECTION 09

S3 Event Notifications

Trigger downstream actions automatically when objects are created, modified, or deleted.

Supported Destinations

- SNS Topics
- SQS Queues
- Lambda Functions
- EventBridge (recommended for advanced filtering)

Event Types

- `s3:ObjectCreated:*` — Put, Post, Copy, CompleteMultipartUpload
- `s3:ObjectRemoved:*` — Delete, DeleteMarkerCreated
- `s3:ObjectRestore:*` — Glacier restore initiated/completed
- `s3:Replication:*` — Replication events
- `s3:LifecycleTransition` — Storage class transition

SECTION 10

S3 Replication

Automatically replicate objects across buckets, within the same region or across different regions.

Replication Types

- **Cross-Region Replication (CRR)** — Different AWS regions. Use cases: compliance, disaster recovery, reduced latency
- **Same-Region Replication (SRR)** — Same AWS region. Use cases: log aggregation, live replication between accounts

Requirements

- Versioning enabled on *both* source and destination buckets
- IAM role with replication permissions
- Optionally: S3 RTC (Replication Time Control) for 99.99% replication within 15 minutes

What Gets Replicated

- New objects after enabling replication
- Metadata and ACLs
- Object tags
- Optionally: delete markers, existing objects (via S3 Batch Replication)

> **WHAT DOES NOT GET REPLICATED**
>
> Objects created before replication was enabled (unless using Batch Replication), objects encrypted with SSE-C, and objects already in Glacier or Deep Archive.

SECTION 11

Data Lake Architecture on S3

S3 is the de-facto foundation for modern data lake implementations on AWS. A well-structured data lake uses clear zone separation.

Modern Data Lake Structure

```
s3://my-data-lake/
├── raw/                          # Unprocessed ingested data
│   └── year=2025/
│       └── month=01/
│           └── day=15/
│               └── data.parquet
├── processed/                    # Cleaned / transformed
│   └── customer_data/
│       └── partition_date=2025-01-15/
│           └── customers.parquet
├── curated/                      # Business-ready analytics
│   └── analytics/
│       └── monthly_reports/
└── archive/                      # Long-term retention
    └── year=2023/
```

Production Bucket Setup — Key Steps

1. Create bucket in the appropriate region
2. Enable versioning on the bucket
3. Enable default encryption (SSE-S3 or SSE-KMS)
4. Block all public access settings
5. Enable server access logging
6. Configure lifecycle policy for zone transitions
7. Configure bucket policy (least-privilege)
8. Add resource tags for cost allocation

SECTION 12

Tips & Best Practices

Security Tips

Tip 1 — Enable Default Encryption and Block Public Access

Always enable both settings for new buckets. Block Public Access should be enabled at the account level to prevent any accidental exposure.

Tip 2 — Use Presigned URLs for Temporary Access

Generate time-limited presigned URLs to grant temporary access to specific objects without making the bucket public.

Tip 3 — Implement MFA Delete for Critical Buckets

Enable versioning first, then enable MFA delete. This requires a valid MFA token for any delete operation, protecting against accidental or malicious deletion. Requires root account with MFA.

Performance Tips

Tip 4 — Use Multipart Upload for Large Files

Always use multipart upload for files > 100 MB. It enables parallel uploads, improves throughput, and allows recovery from partial failures without restarting the entire upload.

Tip 5 — Enable Transfer Acceleration for Global Users

Uploads route through the nearest CloudFront edge location, then traverse the private AWS network to the bucket region. Typical speedup: 50–500% depending on the user's location.

Cost Optimization Tips

Tip 6 — Use S3 Intelligent-Tiering for Unknown Access Patterns

Automatically moves objects between frequent, infrequent, and archive tiers based on access patterns — eliminating the need to predict usage.

Tip 7 — Implement Lifecycle Policies to Reduce Costs

Automate transitions of aging data to cheaper storage classes. Common pattern: Standard → Standard-IA (30d) → Glacier Flexible (90d) → Glacier Deep Archive (180d).

Tip 8 — Monitor and Delete Incomplete Multipart Uploads

Incomplete multipart uploads accrue storage charges. Use a lifecycle rule to automatically abort incomplete multipart uploads after a defined number of days.

Operational Tips

Tip 9 — Use S3 Inventory for Large-Scale Operations

Enable S3 Inventory to generate periodic reports of all objects and their metadata. Query results with Amazon Athena for cost analysis, compliance auditing, and lifecycle verification.

Tip 10 — Implement S3 Event Notifications for Automation

Use event notifications to trigger Lambda functions, SQS queues, or SNS topics in response to object events. Use EventBridge for advanced filtering and routing to multiple targets.

SECTION 13

Pitfalls & Remedies

Pitfall 1: Public Bucket Exposure — Data Security Risk

Problem: Accidentally making S3 buckets or objects public, leading to data breaches.

WHY IT HAPPENS

- Overly permissive bucket policies
- ACLs allowing public access
- Not using Block Public Access settings
- Misconfigured CloudFormation/Terraform

IMPACT

- Data breaches (customer data, credentials, source code)
- Compliance violations (GDPR, HIPAA)
- Reputational damage and financial losses

REMEDY

1. Enable Block Public Access at the account level
2. Audit existing buckets with S3 Storage Lens or AWS Config
3. Fix public buckets — enable block public access, remove public bucket policies
4. For legitimate public content, use CloudFront with OAC instead of public bucket
5. Implement monitoring with CloudWatch Events for policy change alerts

PREVENTION

- Enable Block Public Access at account level
- Use SCPs to prevent disabling Block Public Access
- Conduct regular security audits
- Implement least-privilege IAM policies
- Use CloudFront with OAC for all public content delivery

Pitfall 2: Lifecycle Policy Mistakes — Data Loss / Wasted Cost

Problem: Lifecycle policies unintentionally delete data or fail to reduce costs.

WHY IT HAPPENS

- Incorrect filter configuration
- Not understanding transition constraints
- Overlapping rules with conflicting behavior
- No testing before production deployment

IMPACT

- Accidental data deletion
- Failed transitions (minimum days not met)
- Wasted storage costs (rules not triggering)
- Compliance violations

REMEDY

1. Understand transition constraints (see Section 05)
2. Test lifecycle policies on a small, isolated prefix first
3. Use object tags for fine-grained, per-object control
4. Implement safety checks — versioning + MFA delete before applying deletion policies

PREVENTION

- Test policies on small prefixes first
- Fully understand transition constraints before authoring rules
- Monitor lifecycle actions with CloudWatch Events
- Document policy purpose and expected behavior

Pitfall 3: Cross-Region Replication Issues — DR Failure Risk

Problem: Replication fails, is incomplete, or creates unexpected costs.

WHY IT HAPPENS

- Versioning not enabled on one or both buckets
- Incorrect IAM permissions on the replication role
- Misunderstanding what gets/doesn't get replicated
- Missing encryption compatibility between buckets

IMPACT

- DR strategy silently fails
- Data inconsistency across regions
- Unexpected data transfer costs
- Compliance failures

REMEDY

1. Verify prerequisites: versioning enabled on both buckets, replication IAM role created and policy attached
2. Configure replication rule properly with correct source/destination filters
3. Monitor replication metrics in CloudWatch (ReplicationLatency, BytesPendingReplication)

PREVENTION

- Enable versioning before configuring replication
- Test replication with a small dataset first
- Monitor replication metrics continuously
- Understand what gets/doesn't get replicated (review Section 10)
- Account for costs: data transfer + storage in destination region

SECTION 14

Summary & Key Takeaways

Amazon S3 is the cornerstone of AWS storage, providing durable, scalable, cost-effective object storage for virtually any use case. Understanding storage classes, lifecycle policies, encryption, access controls, versioning, replication, and performance optimization is essential for building production-grade applications on AWS.

- **Choose the appropriate storage class** — Standard for hot data, Intelligent-Tiering for unknown patterns, IA for infrequent access, Glacier for archives, Deep Archive for compliance retention.
- **Implement security layers** — Block Public Access, default encryption, bucket policies, IAM policies, and access points all work together in a defence-in-depth model.
- **Use lifecycle policies** — Automate transitions to cheaper storage classes, expire old data, and clean up incomplete multipart uploads to control costs.
- **Enable versioning** — Protect against accidental deletion, meet compliance requirements. Pair with lifecycle policies to manage version accumulation costs.
- **Optimize performance** — Use multipart uploads for large files, prefix sharding for high throughput, and Transfer Acceleration for global users.
- **Monitor and optimize costs** — Use S3 Storage Lens for visibility, Intelligent-Tiering for automatic optimization, and S3 Inventory for large-scale operational audits.

> **BOTTOM LINE**
>
> Understanding S3 deeply enables you to build cost-effective, secure, high-performance data storage solutions that scale to exabytes. Mastering S3 configuration is one of the highest-leverage skills for any AWS practitioner.